LLM Sampling Methods

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1 Introduction

For LLM output there are lots of decoding methods, each deciding how to pick the next token from the model's output distribution.

Terminology: deterministic search and stochastic sampling are both *decoding* methods. Since beam search is now rarely used, people often say "sampling" to mean the whole decoding category (even in vLLM/SGLang); be aware of the scope when reading.

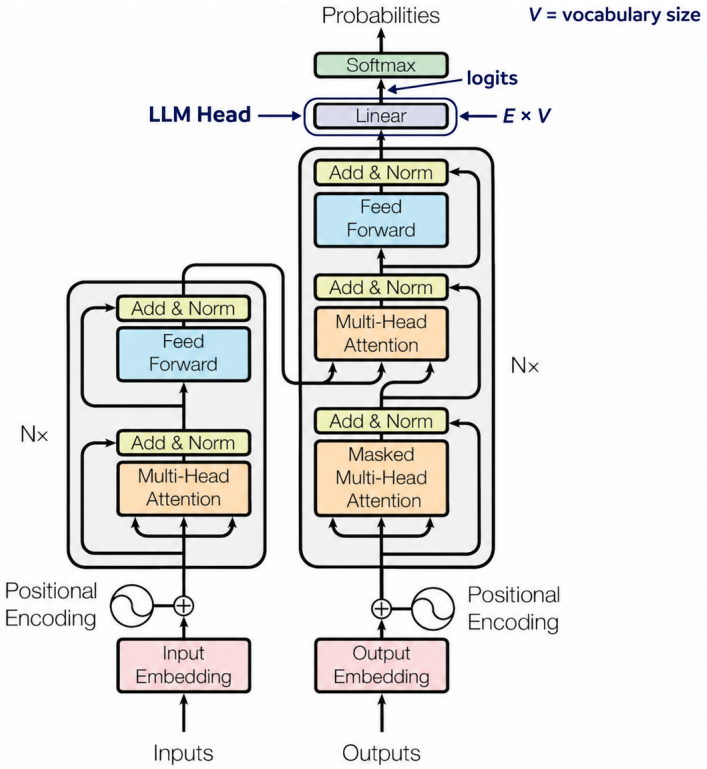


Figure 1: LLM Head Structure

2 Deterministic Search

- **Greedy Search**: always pick the token with the highest probability (arg max only, softmax is not needed since the order unchange), i.e. deterministic decoding. *Drawback:* tend to repeat because losing too much info.
- **Beam Search**: keep the top-N most probable partial sequences, like N beams advancing in parallel through a V-branch tree (V = vocabulary size). *Drawback:* still easily repetive, root case not solved.

3 Stochastic Sampling

- **Pure Sampling**: sample the next token from the full softmax distribution over the entire vocabulary. *Drawback:* the long tail jointly carry non-trivial mass, so pure sampling often yields degenerate text.
- **Top-K Sampling**: keep only the K highest-probability tokens, renormalize, then sample. *Drawback:* fixed rigid K.
- **Top-P (Nucleus) Sampling**: keep the smallest set of tokens whose cumulative probability reaches p, renormalize, then sample; the cutoff adapts to the distribution. *Drawback:* still rigid truncation.
- **Min-P Sampling**: keep tokens with probability $\geq p \times \max(p_i)$, renormalize, then sample. Stabler under high temperature. *Drawback:* don't know how to decide threshold, vLLM/SGLang didn't provide default value till now.

Temperature Factor τ rescales the logits z_i before softmax, $p_i = \frac{\exp(z_i/\tau)}{\sum_j \exp(z_j/\tau)}$, Obviously:

- $\tau < 1$ sharpens the distribution (shrinks the tail; $\tau \to 0$ approaches greedy, which is a common implementation).
- $\tau = 1$ leaves it unchanged.
- $\tau > 1$ flattens it (spreads mass into the tail; $\tau \to \infty$ approaches uniform).

Unlike top-k/top-p, temperature never zeroes out a token — it softly reweights but does not cut the tail.

4 Summary

Top-p sampling needs probabilities *fully sorted* (descending) to accumulate the cutoff; top-k sampling and beam search only need *partial sorted* (top-K selection). Bitonic Sort is a powerful method for these sampling methods.

There are also other decoding type like **Speculative Decoding** for speed, and **Constrained Decoding** for structured output; but they are quite complex and different, need large infrastructure, deserving separate essay to discuss.

References

[1] P. von Platen, *How to generate text: using different decoding methods for language generation with Transformers*, Hugging Face Blog, 2020, huggingface.co/blog/how-to-generate.